Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/30/2018
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Iron Overload and Oxidative Damage: Regulators of Bone Homeostasis in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	College Station	**State:**	TX
Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	12/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2018 per NSSC information (Ed., 6/19/18) NOTE: End date changed to 5/31/2018 per F. Hernandez/ARC (Ed., 7/31/17) NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 5/5/16)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Ford, John Ph.D. (Texas Engineering Experiment Station) Hogan, Harry Ph.D. (Texas Engineering Experiment Station) Turner, Nancy Ph.D. (Texas AgriLife Research)		
Grant/Contract No.:	NNX13AL25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent work in bone biology field has demonstrated that cellular damage due to molecules called reactive oxygen species (so-called oxidative damage) is a key mechanism contributing to aging-related bone loss, as well as to that occurring with the loss of estrogen after menopause. We know very little about whether the loss of gravity as experienced in the space environment modulates oxidative damage to bone. A newly recognized source of oxidative damage in bone is that due to iron overload, or an increase in body iron stores, as has now been demonstrated in both International Space Station (ISS) crew members. Data published this month indicate that sub-pathological levels of serum ferritin (a marker for iron stores) contribute to accelerated bone loss and to incident vertebral fractures in healthy middle-aged men and women. We have preliminary data indicating that similar moderate elevations in body iron stores, as induced by a high iron diet, can result in loss of cancellous bone and diminished bone formation activity in rodents. The experiments in this proposal will utilize iron overload as an experimental intervention to exacerbate oxidative damage during a period of simulated microgravity (by hindlimb unloading in male and female mice) to define the sex-specific mechanistic role of oxidative damage as a key regulator of bone mass and, ultimately, fracture risk as modulated by the gravitational unloading of the space environment. In a final specific aim, we will test whether superimposing acute and continuous (over 42 days) radiation exposure exacerbates oxidative damage and bone decrements during hindlimb unloading. These experiments are deliberately designed to enable a rapid transition to flight experiments requiring minimal intervention during flight aboard ISS, should these ground-based simulations yield provocative findings to justify flight experiments. Iron overload can be maintained by dietary intake, and limiting the acute radiation exposure in Experiment 2 to just before the unloading period allows for testing of the same hypotheses in a flight experiment. The experiments outlined in this proposal address how microgravity affects key regulatory mechanisms governing alterations in bone (from OCB-2 in the Space Biology Plan) and also one of the National Research Council (NRC) Decadal Survey's highest priority recommendations, to expand our understanding of sex differences in adaptation to the space environment, particularly in the area of bone and long-term radiation risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have evidence that oxidative damage caused by molecules that oxidize DNA or other tissue components can contribute to the loss of bone mass and integrity that occurs with aging and with the loss of estrogen in post-menopausal women. There are some unfortunate individuals who experience pathological increases in body iron stores (with diseases like hemochromatosis) who, we believe, also experience increased oxidative damage to multiple tissues due to an excess of free iron molecules. However, we now know that even moderate (sub-clinical) elevations in body iron stores in healthy middle-aged individuals can result in accelerated bone density loss. Only women in these groups, however, exhibit an increased risk (5-fold) of incident vertebral fractures. These experiments in skeletally mature mice will yield important and detailed information on the impact of these moderate elevations in body iron on evidence for oxidative damage and for accelerated bone loss, with the unique added treatment of reduced weight bearing (simulating prolonged bed rest in humans). Because this project will be examining both male and female mice, we will be able to define if there is any sex difference in the response of bone to iron overload, with and without coincident disuse. Our final specific aim will test the impact of increased iron stores in combination with disuse and exposure to chronic, very low dose gamma radiation, simulating space radiation. Radiotherapy patients on Earth experience a 2.6-fold increase in fracture risk compared to age-matched peers. A better understanding of the additional impact of radiation on bone integrity, with and without iron overload, may lead to better protection of bone health in radiotherapy patients, reducing fracture risk.
Task Progress:	Tasks accomplished --Year 4 (no-cost extension year) [NOTE: we received last increment of funding for this project in mid-July 2016, several weeks before initial end date] Finishing final animal experiments [Continuous irradiation (CIRR) in weightbearing and hindlimb unloaded mice, on normal and high Fe diets]: Our final planned experiment (continuous exposure to very low dose rate gamma radiation [CIRR]) will provide unique and valuable data. Three cohorts of mice, each completing 28 days of sham or actual CIRR) were completed by December 15, 2016. To leverage the investment in creating this homogenous radiation field in our animal facility, we have generated a number of piggyback projects with additional cohorts of mice exposed to CIRR before decommissioning the radiation field that continued into 2017, to answer these additional science questions: a. What is the impact of CIRR ± HU in endurance trained mice vs sedentary mice? We added four groups of 10 female mice each who performed regular treadmill running 5 days/week during the 28 days of CIRR ± HU. [November 2016-February 2017; these data will be a core part of R. Bokhari's Ph.D. dissertation] b. Does the impact of 0.5 Gy gamma radiation on bone parameters differ if the radiation dose is delivered in one bolus or if it is delivered continuously at very low dose rates in weightbearing mice? Female and male C57/Bl6 mice, obtained from breeding colony of co-investigator Dr. Nancy Turner, exposed to 70 days CIRR, will be compared to age-matched mice from Dr. Turner's Human Research Program (HRP)-funded protocol investigating radiation impact on intestinal stem cells that were exposed to one single 0.5 Gy dose. [September 2016- June 2017] c. Does the bone response to CIRR change with aging in weightbearing mice? Female C57/Bl6 mice, obtained from breeding colony of co-investigator Nancy Turner, and aged up to ~54 weeks, were exposed to 28 days CIRR and will be compared to our main protocol's weightbearing mice, age ~20 weeks, exposed to 28 days of CIRR. [February- April 2017] d. Are there early biological changes in bone cell activity during exposure to CIRR? Female C57/Bl6 weightbearing mice, 20 weeks of age, will be sacrificed after 4 and 12 days in CIRR (~0.8 mRem/hour) to compare to female mice terminated after 28 d CIRR. [May – August 2017; these data will serve as MS thesis for A. Slavinksy Marich] Tissue analyses have been on-going throughout the year, with significant progress made. Work is on-going to complete these analyses: immunostaining of osteocytes for oxidative damage marker, serum and liver assays for oxidative stress markers and iron status indicators. Interim Results: See Bibliography section for citations to publications with interim results.

Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Bokhari R, Metzger C, Allen M, Lenfest S, Kosniewski J, Seidel D, Hogan H, Turner N, Zwart S, Bloomfield S. "Sex-specific cellular differences in mouse bone with moderate iron elevations leads to differences in mechanical properties." Presented at the ASBMR 2016 (American Society for Bone and Mineral Research) Annual Meeting, Atlanta, Georgia, September 16-19, 2016. Journal of Bone and Mineral Research. 2016;31(Suppl 1):Presentation Number: SU0028. Accessed on 3/30/2018 at http://www.asbmr.org/education/AbstractDetail?aid=e6933838-e8c4-4210-8b2d-630a309f3d16 , Sep-2016
Articles in Peer-reviewed Journals	Macias BR, Lima F, Swift JM, Shirazi-Fard Y, Greene ES, Allen MR, Fluckey J, Hogan HA, Braby L, Wang S, Bloomfield SA. "Simulating the Lunar environment: Partial weightbearing and high-LET radiation induce bone loss and increase sclerostin-positive osteocytes." Radiat Res. 2016 Sep;186(3):254-63. Epub 2016 Aug 18. http://dx.doi.org/10.1667/RR13579.1 ; PubMed PMID: 27538114 , Sep-2016
Articles in Peer-reviewed Journals	Lima F, Swift JM, Greene ES, Allen MR, Cunningham DA, Braby LA, Bloomfield SA. "Exposure to low-dose X-ray radiation alters bone progenitor cells and bone microarchitecture." Radiat Res. 2017 Oct;188(4):433-42. Epub 2017 Aug 3. https://doi.org/10.1667/RR14414.1 ; PubMed PMID: 28771086 , Oct-2017